

British Math Olympiad Round 1 2015 — P3/6

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Suppose that a sequence $t_0, t_1, t_2, \dots$ is defined by a formula $t_n = An^2 + Bn + C$ for all integers $n \geq 0$. Here A, B and C are real constants with $A \neq 0$. Determine values of A, B and C which give the greatest possible number of successive terms of the sequence which are also successive terms of the Fibonacci sequence.

The Fibonacci sequence is defined by $F_0 = 0, F_1 = 1$ and $F_m = F_{m-1} + F_{m-2}$ for $m \geq 2$.

Solution

Suppose that $t_k = F_n, t_{k+1} = F_{n+1}, t_{k+2} = F_{n+2}, \dots, t_{k+q} = F_{n+q}$ for positive integers k, n and $q \geq 2$. Then by the properties of the fibonacci sequence $t_m = t_{m-1} + t_{m-2}$ for any integer $k+2 \leq m \leq k+q$. That is

$$Am^2 + Bm + C = [A(m-1)^2 + B(m-1) + C] + [A(m-2)^2 + B(m-2) + C]$$

but by the Fundamental Theorem of Algebra this equation has at most 2 solutions, and thus there exist at most 2 values of m . That means that the number of integers in the interval $[k+2, k+q]$ is at most 2, and so $q \leq 3$ which tells us that t_n can contain at most 4 consecutive fibonacci terms. Now we present the construction of a quadratic sequence that gives the maximal 4 consecutive fibonacci sequence terms.

Use $t_n = \frac{1}{2}n^2 - \frac{1}{2}n + 2$ and then notice that $t_1 = 2 = F_3, t_2 = 3 = F_4, t_3 = 5 = F_5, t_4 = 8 = F_6$. So the values $A = \frac{1}{2}, B = -\frac{1}{2}, C = 2$ give the greatest possible number of successive terms which are also successive terms of the fibonacci sequence. ■